

3GPP TS 23.501

System architecture for the 5G System (5GS)

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SYNTHETIC SAMPLE DOCUMENT — written for demonstration and testing of a retrieval-augmented chatbot. This is original content structured to resemble a 3GPP technical specification; it is NOT an official 3GPP publication and must not be used as a source of normative requirements.

4.2.1 Network functions overview

The 5G Core network is built from a set of network functions (NFs) that interact through a service-based architecture. The Access and Mobility Management Function (AMF) terminates the NAS signalling from the UE and handles registration, connection, and mobility management. The Session Management Function (SMF) is responsible for session establishment, IP address allocation, and selection of the User Plane Function (UPF) that handles user data forwarding.

4.2.3 Policy and data management functions

The Policy Control Function (PCF) provides policy rules to control-plane functions such as the AMF and SMF, including QoS and charging-related policies. The Unified Data Management (UDM) function stores subscription data and generates authentication credentials in cooperation with the Authentication Server Function (AUSF). The Network Repository Function (NRF) maintains a registry of available NF instances and supports service discovery between them.

5.15.3 Network slicing

Network slicing allows the 5G System to support multiple logical networks, each tailored to a particular set of requirements, over a shared physical infrastructure. A Network Slice is identified by Single Network Slice Selection Assistance Information (S-NSSAI), which comprises a Slice/Service Type (SST) and an optional Slice Differentiator (SD). A UE may be simultaneously served by up to eight Network Slices via a single registration.

5.15.4 Standardized SST values

3GPP defines a small set of standardized Slice/Service Type values so that operators can achieve baseline slice interoperability without prior agreement.

SST value	Slice/Service type	Characteristics
1	eMBB	Enhanced Mobile Broadband — optimised for high data rates
2	URLLC	Ultra-Reliable Low-Latency Communications
3	MIoT	Massive IoT — optimised for high device density, low data rate
4	V2X	Vehicle-to-Everything communications

8.2.1 Registration management states

From a registration management perspective, a UE is in one of two states with respect to the AMF: RM-DEREGISTERED, in which the UE is not registered with the network and the AMF holds no valid location or routing information for it, and RM-REGISTERED, in which the UE has successfully completed a registration procedure and the network can reach it for paging or session-related signalling.